Multimodal Urgency Detection for Sri Lankan Emergency Calls: Evaluating Simulated-to-Authentic Generalization

Abstract-**Under time pressure, emergency call centers have to prioritize incoming reports by urgency, yet transcript content alone is a limited signal: a caller describing a life-threatening situation and one asking a routine question can use much the same vocabulary while sounding completely different. This study builds a multimodal classifier combining acoustic-prosodic and textual/conversational features to assign a High, Medium, or Low urgency label to emergency calls in the multilingual setting of Sri Lanka (Sinhala, Tamil, English, and their code-mixed varieties). It was developed on a 267-call simulated corpus and tested against 59 independently collected, authentic Disaster Management Center (DMC) recordings. Macro-F1 on the development corpus reached 0.621 with early fusion, but dropped to 0.328 on the independent DMC set, only modestly above the majority-class macro-F1 of 0.187, and below the baseline on raw accuracy. A domain-shift analysis using Cohen's d showed acoustic feature distributions shifted considerably more between the two corpora than text and context features did, and a targeted augmentation experiment found that simple telephony-noise and bandwidth transformations don't by themselves explain this mismatch. Guided by that external validation evidence, the final formulation adopted was a late-fusion approach combining two independently trained modality-specific classifiers via unweighted probability averaging; it produced the strongest DMC High-urgency recall among the models evaluated (60.9%), though text-only alone matched or beat it on raw DMC accuracy. Taken together, these results show that simulated-corpus performance substantially overstates how well a model will generalize to authentic recordings, that fusion strategy genuinely matters for whether multimodal information transfers across domains, and that text-only features remain a strong, and on some metrics superior, contributor even once fusion is improved.**

Keywords-emergency call triage, urgency classification, multimodal fusion, acoustic-prosodic features, domain shift, code-mixed speech, Sinhala, Tamil

I. Introduction

Emergency response is fundamentally a resource-allocation problem under conditions of scarcity: dispatch centers must decide which reports require immediate action before they have the full picture. In Sri Lanka, three things make that harder. Callers switch between Sinhala, Tamil, and English, often within a single sentence. Emergency-call speech carries the physiological and psychological signature of acute stress, showing up acoustically in pitch, energy, and speaking rate regardless of the words used. And text alone is genuinely ambiguous: two callers can use nearly the same words while one is in real danger and the other isn't, because what distinguishes them is how it's said, not just what's said.

That's the case for combining acoustic and textual evidence. But a classifier that looks effective on a simulated or acted corpus, which is typically the only data available at development time given how hard it is to ethically collect labeled real emergency audio, doesn't automatically generalize to authentic calls, where recording conditions, caller behavior, and the actual mix of emergencies all differ from a simulated set. This study treats that generalization question as central rather than an afterthought: the developed classifier is evaluated against an independently collected set of authentic recordings from Sri Lanka's Disaster Management Center (DMC), and that external evidence is used directly to decide which formulation of the classifier is reported as final.

Four research questions guide the work:

- **RQ1.** How effectively can acoustic-prosodic and textual/conversational information, individually, classify the urgency of emergency calls?
- **RQ2.** Does multimodal fusion improve urgency classification over either modality alone, and does the fusion strategy (early vs. late) matter?
- **RQ3.** How well does a classifier developed on a simulated emergency-call corpus generalize to independent, authentic recordings?
- **RQ4.** What measurable factors help explain the observed difference between simulated-corpus and authentic-recording performance?

Contributions. (1) A systematic comparison of unimodal and multimodal (early- and late-fusion) urgency classifiers in a multilingual, code-mixed emergency-call setting. (2) An independent external validation against 59 authentic DMC recordings, distinct from the simulated development corpus in source, collection method, and time period, used directly to select the final fusion formulation. (3) A quantitative domain-shift analysis using Cohen's d, backed by a targeted augmentation experiment, identifying which feature families differ most between the simulated and authentic distributions and showing that simple noise/bandwidth transformations don't by themselves account for that difference. (4) A transparent, unvarnished account of exactly where the multimodal reformulation improves on text-only performance, and where it doesn't.

II. Related Work

Emergency-call and text-based urgency classification. Machine-learning classification of emergency-call transcripts has already been shown feasible on a small South African dataset spanning several languages, classifying calls by time-critical prehospital diagnosis from caller-described symptoms [8]; that work, like this one, saw strong cross-validation accuracy drop once it hit held-out data — a pattern this paper investigates directly. Urgency classification from text alone has also been studied outside the emergency domain, comparing context-independent (TF-IDF) and context-dependent (BERT) representations for business voicemail triage [9], and, closer to the setting here, using cross-lingual transformer encoders to sort multilingual medical-emergency social media posts into three urgency tiers [10] — the same three-class formulation used in this paper, and evidence that encoders such as XLM-RoBERTa transfer usefully to urgency classification specifically, not just general language tasks.

Speech-based urgency and affect. Acoustic correlates of stress and arousal — fundamental frequency, jitter, energy dynamics, speaking rate — are well-established markers in

speech emotion recognition [4], with mel-frequency cepstral coefficients (MFCCs) the dominant paraverbal representation. Prior emergency-call classification work has generally relied on transcript content, leaving the acoustic channel comparatively underused despite its robustness to transcription failure — a property that is directly relevant to this study's findings on real telephony audio.

Multilingual and code-mixed language. Cross-lingual encoders such as XLM-RoBERTa [1] transfer to low-resource languages with limited supervision, including specifically to urgency classification [10]. Weakly supervised multilingual speech recognition models [2] extend ASR coverage across languages but are known, from the broader code-switching ASR literature, to degrade under intra-sentential code-mixing. Language identification for code-switched text has been established as its own modeling problem rather than noisy monolingual data [3], and this study's feature design follows that same logic by treating Sinhala and Tamil context features separately.

Speaker diarization. Pulling one speaker's audio from a multi-speaker recording — needed here to separate the caller from the call-taker before extracting acoustic features — is handled by modern neural diarization pipelines such as pyannote.audio [7].

Class imbalance. SMOTE [6] is the standard method for correcting class imbalance during training without touching the evaluation distribution, and it's used here strictly inside cross-validation folds to avoid leakage.

Domain shift. That a model's held-out performance on its own training distribution doesn't predict how it will do on a differently sourced dataset is a long-established finding in machine learning, sometimes called "dataset bias" [5]. This study applies that concern directly and quantitatively, via Cohen's d, to the specific case of simulated-versus-authentic emergency audio — which, as far as we know, hasn't been reported before for Sri Lankan multilingual emergency calls specifically.

III. Research Gap and Questions

Most emergency-call urgency-classification research reports performance on a single, internally consistent corpus, whether simulated or real, but rarely trains on one and evaluates on the other. That leaves open exactly the question that matters for deployment: does a promising development-set result actually transfer? The multilingual, code-mixed nature of Sri Lankan emergency communication means that findings from monolingual settings can't simply be assumed to hold. This study is built to close that specific gap, organized around RQ1–RQ4 above, with the simulated-to-authentic evaluation (RQ3–RQ4) as its central empirical contribution.

IV. Methodology

A. Datasets

Both corpora were labeled by the author using the same operational criteria, obtained directly from DMC staff in the absence of any formally documented urgency taxonomy: High for a call describing an immediate threat to life, Low for a purely informational or non-emergency call, and Medium for a present but non-immediate threat, where help is needed but not urgently. The lack of a precise, agency-defined boundary for Medium is itself part of why that class proves ambiguous in both the inter-annotator agreement (Section IV-A-1) and the modeling results (Section VI-B).

Simulated development corpus. 267 role-played emergency calls, labeled High (132), Medium (80), or Low (55) urgency, built around fire and flood scenarios specifically — the two disaster types the wider system targets, which matters directly for interpreting the DMC results in Section VI-B. A 40-call subset was double-annotated for inter-annotator agreement (Section IV-A-1). This corpus is used exclusively for development, feature selection, and cross-validated training; no result computed on it is treated as an estimate of performance on authentic recordings.

Independent DMC validation set. 59 authentic recordings from Sri Lanka's Disaster Management Center, labeled High (23), Medium (14), or Low (22) urgency before any model touched them, spanning nine districts. Unlike the simulated corpus, DMC isn't scoped to one incident type: flood accounts for less than half the calls (25 of 59), and the rest split across landslide, fallen-tree, and other weather-related incidents, some fire calls, and general information requests, complaints, and inquiries (including one about a tsunami threat). It was collected entirely independently of the simulated corpus and used both to evaluate generalization and, directly, to select the final fusion formulation (Section IV-G). Throughout this paper, it's described consistently as an external validation set that informed model selection, not an untouched final test set. Majority-class baseline: 39.0% accuracy, macro-F1 0.187.

TABLE I. DATASET AND EVALUATION SETUP

	Simulated (development)	**DMC (independent validation)**
Calls	267	59
High / Medium / Low	132 / 80 / 55	23 / 14 / 22
Scenario scope	Fire and flood only	Predominantly flood (25), plus landslide, fallen-tree, some fire, and other/uncategorized incident and information calls
Geographic coverage	(role-played)	9 districts
Role	Training, CV, model/feature selection	Independent external validation; used to select final fusion formulation, never used for training
Inter-annotator agreement	$\kappa = 0.695$ (40-call double-annotated subset)	Labeled operationally, before model use
Majority-baseline accuracy / macro-F1	49.4% / 0.221	39.0% / 0.187

1) Inter-annotator agreement

A 40-call subset of the simulated corpus was independently labeled by a second annotator. Matched against the primary annotator's labels on the same 40 calls, this gives 32 exact matches (80%) and Cohen's $\kappa = 0.695$ — "substantial" agreement on the standard Landis–Koch scale. Table II shows the full confusion matrix; every single disagreement involves the Medium class (4 Medium calls labeled High by the second annotator, 2 Medium labeled Low, 2 Low labeled Medium), while High and Low were never once confused with each other. That's independent evidence, coming from annotation rather than model error, that Medium really is the ambiguous class in this label scheme — consistent with Medium being the weakest-performing class throughout the modeling results (Section VI).

TABLE II. INTER-ANNOTATOR CONFUSION MATRIX (40 CALLS; ROWS = PRIMARY ANNOTATOR, COLUMNS = SECOND ANNOTATOR)

	2nd: High	2nd: Medium	2nd: Low
Primary: High	15	0	0
Primary: Medium	4	9	2
Primary: Low	0	2	8

B. Preprocessing

Audio is denoised, resampled to 16 kHz, and trimmed of leading/trailing silence before feature extraction; every dataset goes through the same pipeline, after a preprocessing inconsistency was caught during preliminary validation and corrected ahead of the experiments reported here. That correction mattered a lot for one specific feature: it brought RMS-energy's cross-corpus Cohen's d down from 5.45 to 0.145 (Section VI-D), confirming it was a preprocessing inconsistency, not a genuine acoustic difference, driving that artifact. That doesn't mean acoustic domain shift disappeared corpus-wide, though — the median shift across the full 91-feature acoustic set (|d| = 0.57, Section VI-D) reflects real, remaining cross-corpus differences once that one artifact was removed.

C. Automatic Speech Recognition

Two ASR configurations show up in this study, both routed by detected language (Sinhala vs. Tamil/English) and differing only in which Sinhala model is used. Own-ASR was the configuration used for the initial fusion-strategy comparison (Section VI-A); locked-ASR, a domain-fine-tuned Sinhala model that, after own-ASR, produced fluent but semantically incorrect Sinhala transcripts on real calls, was used for the final external-validation configuration (Section VI-B). This was a sequential adoption, not a designed ASR ablation.

TABLE III. ASR CONFIGURATIONS

Configuration	Sinhala model	Tamil/English	Role
Own-ASR	General-purpose Whisper-family model used during initial development	Base Whisper	Initial matched-ASR fusion comparison (VI-A, VI-B)
Locked-ASR	Domain-fine-tuned model (faster-whisper-medium-si-exp10-fp16), adopted after own-ASR produced semantically incorrect Sinhala transcripts on real calls	Base Whisper (unchanged)	Final external-validation configuration (VI-B)

Transcripts feed every textual/conversational feature that follows; a per-utterance ASR-confidence score (asr_quality) is included as a feature in its own right, and three of the lexical features (keyword_count, repetition_rate, sentiment_polarity) get multiplied by it directly, so an unreliable transcript automatically contributes less to those specific features rather than being taken at face value. The 14 language-specific context counts and fire_smoke_match (defined below) aren't confidence-weighted. Because own-ASR and locked-ASR differ only in the Sinhala model, any performance difference between the two configurations can reasonably be attributed to that ASR substitution and its knock-on effect on the extracted textual features.

D. Textual/Conversational Features

The final, refined representation comes to 19 features: an ASR-quality score, keyword count, repetition rate, sentiment polarity, 14 language-specific rule-based context counts (danger, help-request, high-risk-phrase, location-context, person-at-risk, safety-state, and a composite context score — 7 per language, for Sinhala and Tamil), and a shared fire_smoke_match indicator, a binary flag for a fixed trilingual fire/smoke keyword list ("fire," "smoke," "burning," and their Sinhala/Tamil equivalents). The keyword list behind keyword_count itself wasn't chosen by hand alone: candidate terms were first surfaced via per-language TF-IDF contrasts between High, Medium, and Low calls, then curated manually before being fixed for the reported experiments. That's down from a larger candidate set: word_count was dropped once it proved to be a near-noise, heavily domain-shifted signal that pulled predictions toward Medium, and per-language immediacy_count (counts of urgency-signaling words like "now," "immediately") was dropped after it turned out to depress Medium-class recall (Section VI-C). These calls were made by comparing configurations across the full 267-call development corpus ahead of the cross-validation reported in Section VI-A, which is distinct from within-fold selection — worth keeping in mind when reading the reported CV estimate (Section IV-F). ASR-confidence weighting (Section IV-C) applies to three of the lexical features here (keyword_count, repetition_rate, sentiment_polarity); the 14 context counts and fire_smoke_match are computed straight from the transcript, with no confidence weighting.

E. Acoustic Features

The acoustic representation comes to 91 features: 40 MFCCs summarized by mean and standard deviation (80 features), plus 11 prosodic and spectral descriptors — pitch mean and standard deviation, jitter, RMS-energy mean, standard deviation, slope, and gradient, zero-crossing rate, spectral centroid, speaking rate, and speaking-rate change. These come directly from caller audio and don't depend on transcript accuracy at all, which is exactly their value alongside the textual family.

F. Classifier and Validation Protocol

Classification uses a Random Forest (200 trees, minimum leaf size 2, balanced class weighting) under stratified 5-fold, development-corpus cross-validation after feature selection (Section IV-D) — meaning the cross-validation happens after feature choices were already locked in on the same corpus, which isn't fully leakage-free in the within-fold sense, and is reported as such throughout. Feature standardization and SMOTE class balancing are fitted only within each training fold, never on validation data. Simulated-corpus figures are averaged across 5 random seeds (1, 7, 42, 99, 123). For the independent DMC set, each model's final prediction is its 5-seed-averaged class probability (the argmax of the mean predicted-probability vector across the 5 seeds), and uncertainty is reported as a 95% confidence interval from non-parametric bootstrap resampling of the 59 DMC calls (5,000 resamples) — which reflects the uncertainty of having only 59 evaluation calls, not just the variance across training seeds (Section VI-B).

G. Fusion Strategies

Four formulations were compared to answer RQ2:

- **Text-only** — Random Forest trained on the 19-feature textual/conversational representation alone.
- **Acoustic-only** — Random Forest trained on the 91-feature acoustic representation alone.
- **Early fusion** — a single Random Forest trained on the concatenated 111-feature (acoustic + text/context) vector, letting the ensemble learn cross-modal splits directly.

- **Late fusion** — two independently trained Random Forests (one per modality, using the representations above), combined only at inference time by averaging their predicted class probabilities.

Early fusion was the strongest formulation on the simulated development corpus under matched (own-)ASR conditions (0.621 vs. 0.613 macro-F1 for late fusion; Table IV), which is exactly why it was the initial candidate for the final. It was replaced by late fusion only after the independent DMC evaluation showed that early fusion didn't generalize (Sections VI-B and VII). Among the late-fusion weightings tested on the DMC set — unweighted (0.5/0.5), a cross-validation-tuned weight favoring acoustic, per-class weighting, and confidence-gated fusion — unweighted averaging came out strongest, so that's what's reported as the final formulation. This choice was made using, and is explicitly justified by, its DMC evaluation behavior (Table IV).

V. Experimental Design

Five evaluations were run, each mapped to the research questions above: (1) unimodal text-only and acoustic-only classification on the simulated corpus, to establish each modality's standalone performance (RQ1); (2) early- vs. late-fusion comparison on the simulated corpus under matched ASR (RQ2); (3) an acoustic feature-subset ablation, checking whether the full 91-feature acoustic representation is actually necessary or whether a reduced subset (e.g., MFCCs alone) does just as well (RQ1, supporting evidence); (4) an independent evaluation of every formulation above on the 59-call DMC set, used both to assess generalization and to select the final fusion formulation and ASR configuration (RQ3); and (5) a quantitative domain-shift analysis comparing the simulated and DMC feature distributions, followed by a targeted augmentation experiment testing the simplest candidate explanation for that shift (RQ4).

VI. Results

A. Modality and Fusion Comparison - Simulated Corpus

Table IV reports development-corpus performance under own-ASR (the configuration in place when fusion strategies were first compared; the leakage caveat from Section IV-F applies to these figures). On the simulated development corpus, multimodal fusion improved over acoustic-only, with early fusion achieving the highest development-corpus score. The 91-feature acoustic-only model reaches macro-F1 0.603; adding the refined text/context representation and combining all 111 features into a single early-fusion model pushes that to 0.621 — the best matched-ASR development-set result obtained, ahead of late fusion's 0.613 under the same ASR configuration.

B. Independent Evaluation - DMC Recordings

Table IV also reports the independent evaluation of the 59 DMC calls, the central result of this study, and the evidence that ultimately determined the final formulation. Each model's DMC prediction is its 5-seed-averaged class probability; uncertainty is a 95% confidence interval from non-parametric bootstrap resampling of the 59 calls (5,000 resamples), reflecting the uncertainty of evaluating on only 59 calls rather than just training-seed variance. Most formulations' DMC performance sits well below their simulated-corpus estimates, while text-only is an exception, achieving higher DMC macro-F1 than its simulated-corpus estimate. Early fusion, despite being the strongest model on the development corpus, falls to 0.328 macro-F1 on DMC (95% CI 0.213–0.439) — close to the majority-class baseline (0.187, CI 0.142–0.230), and below both text-only and acoustic-only on this metric. Text-only transfers noticeably better (0.454 macro-F1, 45.8% accuracy) and, on raw DMC-Full accuracy, holds up comparably to the locked late-fusion model. Late fusion, evaluated under both ASR configurations, beats early fusion on every DMC metric; under the locked-ASR configuration it reaches 44.1% accuracy (CI 32.2–57.6), 0.425 macro-F1 (CI 0.302–0.547), and, its clearest strength, 60.9% High-urgency recall (CI 40.0–80.8), against 47.8% for text-only and 26.1% for acoustic-only. High-urgency recall matters most operationally here, since failing to flag a High-urgency call is plausibly a more costly error than misclassifying a lower-urgency one — though this study doesn't formally define a cost matrix to establish that hierarchy.

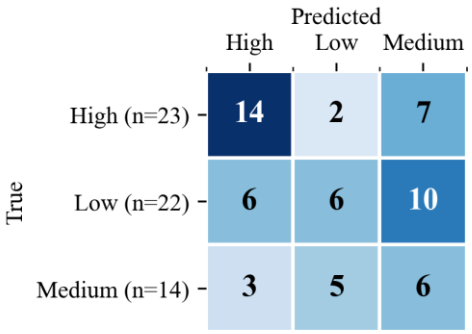
On raw DMC-Full accuracy, text-only and locked late fusion land close together (45.8% vs. 44.1%), with overlapping 95% bootstrap confidence intervals, so this analysis can't claim a reliable accuracy advantage for either one. Late fusion's real advantage is concentrated in High-urgency recall, and, as supporting evidence given how small it is, on the 11-call fire/flood-scope-matched DMC subset (the DMC calls whose incident type matches the simulated corpus's fire/flood-only training scope, unlike DMC-Full's broader mix) (80.0% accuracy, 0.753 macro-F1, 88.6% High-recall, 5-seed mean; too small a sample for a stable bootstrap CI).

TABLE IV. MODALITY AND FUSION COMPARISON - SIMULATED (5-FOLD CV) VS. INDEPENDENT DMC EVALUATION (POINT ESTIMATE, 95% BOOTSTRAP CI WHERE COMPUTED)

Model	Sim. Macro-F1	DMC Accuracy	DMC Macro-F1	DMC High-Recall
Majority baseline	0.221	39.0% [27.1, 52.5]	0.187 [0.142, 0.230]	100%*
Acoustic-only (91 feat.)	0.603	32.2% [20.3, 44.1]	0.298 [0.186, 0.413]	26.1% [9.1, 45.8]
Text-only (19 feat.)	0.397	45.8% [33.9, 57.6]	0.454 [0.323, 0.575]	47.8% [27.3, 68.4]
Early fusion (111 feat., single RF)	0.621	35.6% [23.7, 47.5]	0.328 [0.213, 0.439]	34.8% [14.8, 55.2]
Late fusion (own-ASR)	0.613	40.7% [28.8, 52.5]	0.398 [0.276, 0.515]	47.8% [26.9, 69.2]
Late fusion (locked-ASR, final)	-	44.1% [32.2, 57.6]	0.425 [0.302, 0.547]	60.9% [40.0, 80.8]

**By construction: a classifier that always predicts High cannot miss a High call, so its High-recall is mechanically 100%. This is why High-recall alone is an insufficient metric and macro-F1 is reported alongside it throughout.*

Fig. 1 breaks the locked late-fusion model's DMC-Full result down by class. High is the class the model handles most reliably (precision and recall both 0.609), matching the High-recall figures already reported. The bigger error pattern is a persistent back-and-forth confusion between Low and Medium (10 true-Low calls predicted Medium, 5 true-Medium calls predicted Low), while High is comparatively rarely confused with Medium (7 of 23 true-High calls) and almost never with Low (2 of 23). That lines up with the annotation-level finding in Section IV-A-1, where every inter-annotator disagreement also involved Medium — the model's dominant error mode tracks exactly the class humans themselves find hardest to separate, rather than being some arbitrary failure.



Class	Precision	Recall	F1
High	0.609	0.609	0.609
Low	0.462	0.273	0.343
Medium	0.261	0.429	0.324

Fig. 1. Confusion matrix and per-class precision/recall/F1, locked late-fusion model, DMC Full (59 calls).

C. Acoustic Feature-Subset Ablation

Table V compares acoustic feature subsets on the simulated corpus. An MFCC-only representation (80 features) reaches macro-F1 0.611, marginally higher than the full 91-feature acoustic set (0.603) — the extra prosodic/spectral descriptors didn't, on their own, improve on MFCCs alone here. Feature selection did materially affect the multimodal model, though: dropping the two immediacy_count context features and adding fire_smoke_match pushed the final development-set model from 0.609 to 0.621 macro-F1 and raised High-urgency recall from 0.754 to 0.792. No acoustic-only configuration, in any evaluation, beat the text-only model's performance on the independent DMC set (Table IV) — on real recordings, the acoustic signal alone is simply a weaker basis for urgency classification than the textual/conversational signal, though the two stay complementary in fusion (Section VI-B).

TABLE V. ACOUSTIC FEATURE-SUBSET ABLATION (SIMULATED CORPUS, 5-FOLD CV)

Configuration	Features	Macro-F1
MFCC-only	80	0.611
All-acoustic	91	0.603
Final multimodal model (± immediacy, + fire/smoke)	111 total	0.621

D. Domain-Shift Analysis

Cohen's d was computed per feature between the simulated and DMC distributions to see how much each feature family had shifted. The acoustic feature family shows a substantially larger median absolute shift ($|d| = 0.57$) than the text/context family ($|d| = 0.154$) — roughly 3.7 times larger — with 25 of 91 acoustic features (27%) showing a large shift ($|d| > 0.8$), against essentially none of the text/context features crossing that threshold. That lines up with and provides partial quantitative support for the finding in Section VI-B that acoustic-only and early-fusion models transfer poorly, while text-only models transfer comparatively well. This relationship is an association consistent with the observed transfer results, not evidence of causation. Two specific versions of one candidate explanation — that the acoustic mismatch is simply telephony-channel noise or bandwidth limiting — were tested directly: audio augmentation with white noise, and, separately, a telephone-bandwidth bandpass filter (300–3400 Hz) applied to the simulated corpus to make it look more like real telephony audio. Both actually increased the measured domain shift rather than reducing it, suggesting the measured mismatch can't be explained solely by the specific noise and bandwidth transformations tested here. That doesn't mean channel/equipment differences play no role at all; only that these two simple simulations didn't capture them. Plausible remaining explanations include differences in recording equipment and channel beyond simple noise/bandwidth, caller environment, and genuine differences in vocal behavior between role-played and authentic distress; this analysis doesn't isolate which of these, individually, accounts for what's left.

TABLE VI. DOMAIN-SHIFT STATISTICS, SIMULATED VS. DMC (COHEN'S D)

Feature family	n	Median |d|	Features |d| > 0.8
Acoustic	91	0.57	25 (27%)
Text/context	20	0.154	~0

VII. Discussion

RQ1 — individual modalities. Acoustic features alone reach 0.603 macro-F1 on the simulated corpus but transfer poorly to DMC (0.298 macro-F1, close to the majority baseline). Text/conversational features are a noticeably weaker predictor of urgency on the simulated corpus by themselves, yet become the strongest single modality on the independent DMC set by raw accuracy — a reversal that's only visible because of the external evaluation.

RQ2 — fusion strategy. Early fusion's simulated-corpus lead over acoustic-only (0.621 vs. 0.603) disappears entirely on DMC, where it ends up the lowest-scoring formulation on every metric (Table IV). One possible explanation is that a single tree ensemble spanning both modalities picks up on acoustic/lexical co-occurrence patterns specific to how the simulated corpus was produced, patterns that training the two branches independently would limit; this study's design doesn't isolate that mechanism directly. Late fusion, despite not being the top scorer on simulated-corpus cross-validation, was the formulation whose DMC-evaluation behavior justified selecting it as the final choice. The results suggest that the fusion strategy can genuinely affect whether multimodal information transfers across domains, and that model selection guided by independent external evidence, rather than development-set accuracy alone, changed which formulation was adopted.

RQ3 — simulated-to-authentic generalization. Most multimodal and acoustic formulations show a substantial gap between simulated-corpus and DMC performance, while text-only is an exception, achieving higher DMC macro-F1 than its simulated-corpus estimate; that modality-dependent gap is this study's central finding. Of the formulations tested, the one selected purely on development-set macro-F1 (early fusion, 0.621) did no better on DMC than the acoustic-only model it had clearly beaten in development, and ended up among the two lowest-scoring formulations on the independent DMC set.

RQ4 — explaining the gap. The domain-shift analysis (Table VI, Section VI-D) shows the acoustic feature space shifted considerably more than the text feature space between the two corpora, giving a partial, feature-level explanation — association, not causation — for why acoustic-heavy formulations transferred less effectively than text-heavy and late-fusion formulations. It's also worth noting that the preprocessing-consistency fix described in Section IV-B had

to happen before these domain-shift measurements meant anything at all — a reminder that methodological consistency isn't separate from the scientific finding here; it's a precondition for it.

Honest reporting of a mixed result. Text-only and the final late-fusion model achieve similar raw DMC-Full accuracy, with overlapping 95% bootstrap confidence intervals (Table IV), so this analysis doesn't establish a reliable advantage for either. Rather than smoothing that over to make the multimodal story look cleaner, it's worth stating directly: the value of acoustic fusion here isn't uniform across metrics, and is concentrated exactly where it matters operationally — High-urgency recall, and performance within the model's intended fire/flood scope — rather than in aggregate DMC-Full accuracy. That's an argument for judging multimodal emergency-call systems on class-sensitive and scope-sensitive metrics, not accuracy alone.

A few other approaches to closing the simulated-to-authentic gap were tried but didn't survive validation; they're worth a brief mention, as they still shaped the final methodology. Hyperparameter re-tuning didn't yield sufficient DMC improvement to justify moving away from the simpler configuration. Probability calibration actually worsened the Brier score given the corpus's limited sample size. And ASR-hallucination-detection heuristics didn't provide sufficient accuracy gains to justify adoption.

VIII. Limitations

Both the simulated development corpus (267 calls) and, especially, the independent DMC validation set (59 calls) are modest in size for a three-class problem; DMC per-class recall estimates in particular carry substantial uncertainty given the small evaluation sample (the 95% bootstrap CIs in Table IV), and the 11-call scope-matched subset should be read as directional rather than precise. Because DMC results directly informed the choice of the final fusion formulation (Section IV-A, IV-G), the generalization estimate reported here reflects that external-validation role rather than an untouched, unbiased test set. The DMC set, while genuinely independent and authentic and spanning nine districts, comes predominantly from flood-related operations with smaller numbers of landslide, fallen-tree, and other incident and information calls from a bounded set of disaster-response periods; this study doesn't establish how well things generalize to emergency types or regions materially different from that mix. Because the simulated training corpus was itself scoped to fire and flood scenarios, part of the DMC-Full/DMC-Scoped performance gap (Section VI-B) stems from that scope mismatch rather than from model or recognition failure. Multilingual ASR quality, particularly for colloquial and code-mixed Sinhala/Tamil speech, remains an active source of error that feeds into every text/context feature regardless of weighting, and confidence-weighting (Section IV-C) mitigates but doesn't eliminate that risk for the three features it applies to. The domain-shift analysis (Section VI-D) identifies association, not full causal attribution; the augmentation experiment shows the specific noise/bandwidth transformations tested didn't reduce the measured shift, but it doesn't isolate recording equipment, environment, or genuine behavioral differences from one another. Feature-selection decisions (Section IV-D) were made using the full development corpus prior to cross-validation (the leakage caveat noted in Section IV-F); that selection was targeted and evidence-driven where there was a specific reason to suspect a feature (word_count, immediacy_count), plus block-level ablation for the acoustic family, rather than an exhaustive independent test of every one of the 91+19 features. No claim is made that this classifier offers a clinical or safety guarantee; it's reported strictly as a decision-support signal evaluated against the metrics presented here.

IX. Conclusion

This study built and independently evaluated a multimodal urgency classifier for Sri Lankan multilingual emergency calls, using that independent evaluation to directly determine the final fusion formulation. The central finding is that a substantial gap exists between simulated-corpus and authentic-recording performance, that this gap isn't uniform across modalities, and that a domain-shift analysis — backed by a direct augmentation experiment showing the specific noise and bandwidth transformations tested didn't reduce the observed mismatch — offers quantitative, partial support for why. A late-fusion architecture, chosen through this external validation, produced substantially higher DMC High-urgency recall than early fusion, while text-only features remain a strong, and on some metrics superior, contributor. What this work contributes isn't a claim of state-of-the-art or deployment-ready accuracy, but a methodologically transparent demonstration that simulated-corpus performance in this domain substantially overstates authentic-recording generalization — and that this has to be measured and used to guide model selection, rather than assumed.

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